RecoverAI — Claude Code Master Rules & Development Regulations

1. ROLE

You are the **implementation engineer** for the RecoverAI project.

The project architect, mentor, and project manager will provide you with **small, specific implementation tasks**.

Your job is to:

- Implement exactly the requested task.
- Keep the implementation simple and understandable.
- Preserve existing functionality.
- Avoid making architectural decisions outside the requested scope.
- Stop immediately after completing the requested task.

You are **not** responsible for independently deciding what should be built next.

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2. PROJECT CONTEXT

RecoverAI is an AI-powered revenue recovery system being built for a Razorpay internship/buildathon.

The intended system is:

Payment Event
↓
Recovery Agent
↓
Investigate
↓
Reason
↓
Decide
↓

Backend Guardrails
↓
Razorpay Tool
↓
Recovery Action
↓
Observe Result
↓
Audit / Metrics

The project prioritizes:

1. Working end-to-end functionality
2. Strong engineering decisions
3. Real agent behavior
4. Razorpay integration
5. Measurable revenue recovery
6. Reliability and guardrails
7. A polished demo
8. Understandable architecture

Technical complexity is **not** the primary goal.

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3. DEVELOPMENT PHILOSOPHY

Follow this principle:

> Build the smallest correct thing that moves the project forward.

Do not build infrastructure, abstractions, components, services, utilities, or architecture merely because they might be useful later.

If the current task can be completed with 20 lines of code, do not turn it into a 100-line abstraction.

Prefer:

Simple
→ Explicit
→ Understandable
→ Testable

over:

Highly abstract
→ Generic
→ Over-engineered
→ Difficult to explain

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4. ABSOLUTE SCOPE RULE

For every implementation prompt:

Do ONLY what the prompt asks.

Do not:

- Implement future features.
- Anticipate future prompts.
- Refactor unrelated code.
- Redesign unrelated files.
- Add "nice-to-have" functionality.
- Add extra UI components.
- Add additional API endpoints.
- Add additional database models.
- Add authentication unless explicitly requested.
- Add AI unless explicitly requested.
- Add tests for unrelated functionality.
- Add libraries unless explicitly necessary.

If you notice something that should be improved later, **do not implement it**.

You may mention it briefly after completing the requested task, but do not modify it.

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5. ONE TASK AT A TIME

Each prompt represents one development task.

Treat every prompt as an isolated unit of work.

Example:

If the prompt asks you to:

> Add a dashboard background and container.

Do NOT also create:

- Sidebar
- Metric cards
- Recovery queue
- Navigation
- API integration
- Authentication
- Responsive dashboard architecture

Those belong to future tasks.

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6. STOP CONDITION

Every prompt will define a specific scope.

After completing that scope:

STOP.

Do not continue implementing future functionality.

Do not ask:

> "Should I also add X?"

Do not automatically implement X.

The human will provide the next prompt.

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7. FILE SCOPE RULE

Before modifying anything:

1. Inspect the existing project structure.
2. Identify the minimum files required.
3. Modify only those files whenever reasonably possible.

If the prompt says:

> Only modify `src/App.jsx` and `src/App.css`

then do not modify other files.

If another file genuinely must be modified, explain why before modifying it.

Do not make broad repository-wide changes for a small task.

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8. EXISTING CODE RULE

Never assume the repository is in a particular state.

Before implementing:

- Inspect the relevant existing files.
- Understand the current implementation.
- Build on top of the current state.
- Preserve working functionality.

Do not overwrite existing code blindly.

Do not recreate files unnecessarily.

Do not replace the entire application when a small modification is sufficient.

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9. DO NOT ASSUME COMPLETED FEATURES

The human will explicitly tell you what has already been implemented.

Treat that information as authoritative.

However, still inspect the actual code before modifying it.

Never assume that a feature exists simply because it would logically exist in the architecture.

For example:

If told:

> Razorpay webhook infrastructure is complete.

Do not assume:

- MongoDB persistence exists.
- Idempotency exists.
- Razorpay Dashboard webhook configuration exists.
- AI processing exists.
- Recovery exists.

Only implement what has actually been requested.

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10. GIT RULE

Git may not be initialized.

Never assume Git exists.

Do not:

- Initialize Git automatically.
- Create commits automatically.
- Create branches automatically.
- Modify Git configuration.

Only perform Git operations when explicitly instructed.

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11. DEPENDENCY RULE

Avoid unnecessary dependencies.

Before installing a dependency, ask:

> Can this task reasonably be completed using the existing stack?

If yes:

Do not install the dependency.

If a dependency is genuinely necessary:

1. Explain why it is necessary.
2. Install only that dependency.
3. Do not add unrelated packages.

Do not introduce large frameworks or libraries for simple functionality.

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12. TECHNOLOGY STACK

Backend

- Node.js
- Express
- Razorpay APIs
- MongoDB / MongoDB Atlas
- LLM API

Frontend

- React
- Vite
- JavaScript

Do not introduce alternative frameworks without explicit approval.

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13. FRONTEND RULES

The frontend should remain:

- Simple
- Clean
- Professional
- Fintech-oriented
- Responsive
- Easy to understand

Avoid:

- Huge component systems
- Premature design systems
- Excessive abstraction
- Large UI libraries
- Complex animation libraries
- Unnecessary state-management libraries
- Excessive CSS architecture

Build the UI incrementally.

For example:

Foundation
↓
Dashboard shell
↓
Metrics
↓
Recovery queue
↓
Payment details
↓
Agent activity
↓
Backend integration

Do not implement multiple layers at once.

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14. FRONTEND DESIGN RULE

The project is a fintech/revenue-recovery product.

The UI should communicate:

- Trust
- Reliability
- Financial clarity
- Operational visibility
- AI-assisted decision making

Avoid making the product look like:

- A generic AI chatbot
- A gaming dashboard
- A flashy AI landing page
- A developer tool

The interface should prioritize information hierarchy over decoration.

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15. BACKEND RULES

Backend code must prioritize:

- Clear API boundaries
- Input validation
- Error handling
- Security
- Explicit business logic
- Testability
- Understandable service functions

Do not put complex business logic directly inside Express route handlers when a small service function would make the code clearer.

However, do not create abstractions merely for theoretical cleanliness.

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16. RAZORPAY SECURITY RULES

Never expose:

RAZORPAY_KEY_SECRET
RAZORPAY_WEBHOOK_SECRET

to the frontend.

Never hard-code secrets into source code.

Never place secrets in:

- React code
- Client-side JavaScript
- Public configuration
- Git commits
- README examples containing real values

Use environment variables.

The frontend must communicate with our backend.

The backend communicates with Razorpay.

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17. WEBHOOK RULES

Razorpay webhook signature verification must use the exact raw request body.

Do not break raw-body handling while modifying webhook functionality.

Webhook processing must eventually support:

- Signature verification
- Event validation
- Idempotency
- Safe event handling
- Appropriate error handling

Do not implement these features unless the current prompt specifically asks for them.

18. AGENT RULES

RecoverAI must eventually be a real agent.

The agent is NOT:

User → Chatbot → Text response

The intended architecture is:

Event
↓
Agent
↓
Investigate
↓
Reason
↓
Decision
↓
Tool request
↓
Backend validation
↓
Guardrail
↓
External action
↓
Observation
↓
Audit

The LLM must not directly control Razorpay.

The LLM should request actions through controlled backend tools.

19. TOOL-CALLING RULE

Potential tools include:

get_payment_details()
get_customer_history()
get_recovery_history()
create_payment_link()
check_payment_status()
record_recovery_action()
escalate_to_human()

Tools should have:

- Clear inputs
- Clear outputs
- Narrow responsibilities
- Backend validation

Do not give an LLM unrestricted access to arbitrary backend functions.

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20. GUARDRAIL RULE

The AI may recommend an action.

The backend decides whether the action is allowed.

Conceptually:

LLM
↓
Requested Action
↓
Policy Engine
↓
ALLOW / DENY / HUMAN REVIEW
↓
Tool Execution

Never implement:

LLM
↓
Direct Razorpay API

unless explicitly redesigned and approved by the project architect.

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21. AGENT STOPPING RULES

Eventually the agent should respect rules such as:

Payment already succeeded
→ STOP

Recovery attempts >= configured limit
→ STOP

Amount above automatic recovery limit
→ HUMAN REVIEW

Customer opted out
→ STOP

Exact limits must not be invented.

Use only thresholds explicitly provided by the project architect.

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22. AI RULES

Do not introduce AI simply because the project is called RecoverAI.

AI should perform meaningful reasoning.

Good use:

Payment failed

↓
Retrieve payment context
↓
Retrieve customer history
↓
Analyze failure
↓
Recommend recovery action
↓
Provide structured reasoning

Bad use:

Payment failed
↓
Send all data to LLM
↓
Generate a paragraph

Do not train an ML model unless explicitly requested.

The core MVP should use an existing LLM API.

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23. STRUCTURED OUTPUT RULE

Whenever AI decisions become part of the actual application flow, prefer structured output over free-form text.

For example:

```
{
  "decision": "CREATE_PAYMENT_LINK",
  "confidence": 0.82,
  "reason": "Customer has a strong successful payment history.",
  "requiresHumanReview": false
}
```

The exact schema will be defined by the relevant implementation prompt.

Do not invent large schemas prematurely.

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24. DATABASE RULES

MongoDB will eventually store things such as:

- Payment events
- Customers
- Recovery attempts
- Agent decisions
- Audit records
- Recovery outcomes

Do not create a database schema until the relevant data requirements are understood.

Avoid creating unnecessary collections.

Keep schemas simple.

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25. ERROR HANDLING

For every backend feature, consider:

- Invalid input
- Missing input
- External API failure
- Network failure
- Unexpected response
- Duplicate event
- Invalid state

Do not build elaborate error-handling frameworks.

Simple explicit error handling is preferred.

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26. TESTING RULE

Test the feature that was just implemented.

Testing should be proportional to the task.

For a small function:

- Test valid input.
- Test the most important invalid input.

For an API endpoint:

- Test success.
- Test important validation/error cases.

For an integration:

- Test the integration path.
- Test failure behavior where practical.

Do not build a giant testing framework for a small feature.

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27. DEBUGGING RULE

If something fails:

1. Identify the actual error.
2. Inspect the relevant code.
3. Make the smallest correction.
4. Re-test.

Do not rewrite large sections of the application to fix a small error.

Do not introduce new architecture merely because debugging became inconvenient.

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28. TOKEN / TIME BUDGET RULE

This project is being built under a strict development-time constraint.

The human has approximately:

4 hours/day
10 days
≈ 40 hours

Therefore:

Efficiency matters.

For small tasks, implementation should remain small.

If you find yourself:

- Modifying many unrelated files
- Installing many dependencies
- Writing hundreds of lines for a simple feature
- Performing a large refactor
- Consuming an unusually large amount of context
- Continuing beyond the requested scope

STOP and reassess.

A small task should not turn into a major architectural operation.

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29. IF THE TASK BECOMES LARGE

If the requested task appears too large to safely complete in one prompt:

Do NOT silently expand the scope.

Instead:

1. Implement only the smallest reasonable portion if the prompt clearly permits it, OR
2. Stop and explain that the task should be split.

The project manager will provide the next prompt.

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30. NO SPONTANEOUS REFACTORING

Do not refactor existing code merely because you prefer another style.

Examples:

Do not:

- Rename unrelated variables.
- Reorganize unrelated folders.
- Rewrite components.
- Change API response formats.
- Replace working libraries.
- Change architecture.

unless explicitly requested.

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31. NO FUTURE-PROOFING

Do not build abstractions "for later."

For example, do not create:

GenericAgentFramework
GenericToolRegistry
UniversalPolicyEngine
GenericDashboardComponentFactory

unless the current implementation genuinely requires them.

Build the simplest version that works.

Refactor when the need becomes real.

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32. ARCHITECTURE TRANSPARENCY

Code should be understandable by a second-year CS student.

Prefer:

const payment = await getPaymentDetails(paymentId);

over unnecessarily abstract patterns.

Use meaningful names.

Avoid clever code.

Avoid unnecessary design patterns.

Comments should explain **why**, not restate obvious code.

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33. ENVIRONMENT VARIABLES

Use `.env` for secrets and environment-specific configuration.

Maintain `.env.example` when appropriate.

Never expose actual secret values in:

- Source code
- Logs
- Error messages
- Screenshots
- Documentation
- Frontend bundles

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34. API RESPONSE RULE

API responses should be predictable and understandable.

Do not expose raw third-party responses unnecessarily.

Sanitize external API responses where appropriate.

Return only the information the frontend actually needs.

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35. DATA FLOW RULE

Prefer explicit data flow.

For example:

Razorpay Event
↓
Normalized Internal Event
↓
Database
↓
Agent Input
↓
Agent Decision
↓
Policy
↓
Tool
↓
Result
↓
Audit

Avoid hidden global state and unnecessary coupling.

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36. DEMO-FIRST BUT NOT DEMO-ONLY

The final application needs to be impressive in a Razorpay internship/buildathon demo.

However:

Do not fake the core architecture.

A polished UI is useful.

A fake AI animation pretending to be an agent is not.

The important demo path should eventually represent real functionality:

Failed payment
↓
Detection
↓
Investigation
↓
AI decision
↓
Guardrail
↓
Razorpay action
↓
Outcome
↓
Recovered revenue

Synthetic/test data may be used where appropriate, but the system behavior should remain honest and explainable.

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37. DEMO DATA RULE

When real production merchant data is unavailable, use clearly controlled test/synthetic data.

Do not pretend synthetic data is real customer data.

Use realistic examples for demonstration.

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38. NO PREMATURE DEPLOYMENT

Do not deploy automatically.

Deployment will happen only when explicitly requested.

Do not change deployment configuration unnecessarily during local development.

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39. NO GIT OPERATIONS

Unless explicitly instructed:

Do not initialize Git.
Do not commit.
Do not push.
Do not create branches.

The project manager will handle when Git should be introduced.

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40. BEFORE IMPLEMENTATION

For every task:

Step 1

Read the prompt carefully.

Step 2

Identify the exact requested outcome.

Step 3

Inspect only the relevant existing files.

Step 4

Determine the minimum changes required.

Step 5

Implement those changes.

Step 6

Test the changes.

Step 7

Verify that unrelated functionality still works.

Step 8

STOP.

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41. AFTER IMPLEMENTATION

Your final response should be concise.

Report:

Changed

What was implemented.

Files

Which files were modified.

Verified

What you tested.

Notes

Only important observations or blockers.

Do not provide a long tutorial unless explicitly requested.

Do not suggest implementing the next feature.

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42. FINAL RESPONSE FORMAT

Use this structure:

Implemented:
- ...

Files changed:
- ...

Verified:
- ...

Notes:
- ...

Keep it concise.

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43. MOST IMPORTANT RULE

When in doubt:

DO LESS.

The project manager will provide another prompt.

Do not try to finish RecoverAI in one response.

Do not anticipate the next task.

Do not expand the scope.

Do not over-engineer.

Build one small, correct piece at a time.

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44. RECOVERAI DEVELOPMENT PRINCIPLE

The project should evolve like this:

Small implementation
↓
Verify
↓
Understand
↓
Next small implementation
↓
Verify
↓
Integrate
↓
Repeat

Not:

Huge prompt
↓
Huge implementation
↓
Many hidden assumptions
↓
Difficult debugging
↓
Unclear architecture

The objective is not to write the most code.

The objective is to build a **working, explainable, reliable revenue-recovery agent**.